Lightpanda: A Headless Browser That Never Draws the Page

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2026-09-29

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Lightpanda is a headless browser written from scratch for machines rather than screens, aimed at teams running web scraping and AI agents at volume. It is not a Chromium fork or a WebKit patch: it is one binary written in Zig that keeps the parts automation reads and throws away the part that draws pixels. Pages are fetched over libcurl, parsed with html5ever from the Servo project, and run in V8, Chrome's own JavaScript engine, against an in-memory DOM. It

still speaks Chrome's DevTools Protocol, so existing Puppeteer and Playwright scripts point at it unchanged. In the project's crawler benchmark, 25 parallel workers crawled a 933-link demo store in 4.81 seconds using 123MB; headless Chrome took 46.70 seconds and 2.0GB.

One load-and-extract cycle over the same kind of product page averaged 16 milliseconds against Chrome's 185, at 21MB of peak memory against 402MB, so the same job fits on far fewer machines. Because the built-in agent runs inside the browser process, a finished session can be saved as PandaScript, plain JavaScript that replays with no model involved; after that, every replay is token-free.

About 82% of the Web Platform Tests suite passes, screenshots return placeholders instead of pixels, and in one agent benchmark text-only output lost a few points of accuracy where a rendered page would have shown more. Its licence blocks embedding it in a closed product, and telemetry is on by default.

Read More: PinchTab takes the opposite approach, wrapping a full Chrome session instead of replacing the engine.

Sources:

- Lightpanda browser repository
- Benchmarks and methodology
- Architecture overview
- Introducing Lightpanda Agent and PandaScript
- Web Platform Tests dashboard

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